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TOPOLOGICAL GROUP STRUCTURES IN INTEGER LATTICE SPACES OF ALL DIMENSIONS

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ABSTRACT. Integer lattice spaces are not as amorphous as might be expected. Using a number theoretic approach with geometric interpretation we provide a meaningful decomposition of hyperspheres and identify natural subgroups of $\mathbb{Z}^k$ for $k \in \mathbb{N}$ under certain operations that preserve point topology.

INTRODUCTION

Although this paper uses number theoretic concepts of sums of squares of integers and partitions of numbers it is not about these. Rather, we apply such to topological group structures in k-dimensional integer lattice spaces, elucidating non-trivial subgroups of $\mathcal{E}_k$ based on transformations of sets of lattice points of invariant point topology.

The spaces $\mathbb{Z}^k$, $k \in \mathbb{N}$, are characterised by sets of k integer valued Cartesian co-ordinates $(x_1, \cdots, x_k)$ and in themselves do not appear to be very exciting. But by distinguishing a lattice point as the origin of co-ordinates we find significant inherent structure in sets of lattice points about the origin. We will refer to such k-dimensional spaces of spherically symmetric sets of lattice points as $\mathcal{E}_k$.

1. $\mathcal{E}_k$ AS EUCLIDEAN SPACES

Euclidean vector spaces possess properties of orthogonality and distance defined via vector products. In $\mathcal{E}_k$ spaces the implication is points defined by integer co-ordinates $(x_1, \cdots, x_k)$ in simple k-cubic integer lattices with separation distances between lattice points measured via Pythagorean calculation according to elementary Cartesian geometry. We will however avoid the necessity for square roots by use of the definitely positive form of the square of Euclidean distance in $\mathcal{E}_k$; this

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measure, following Wildberger[1], we will term *quadrance*. Quadrance here is then inherently integer valued. Thus radial measure is given by $|r|^2 = x_1^2 + \cdots + x_k^2$ which is the quadrance of the point $(x_1, \cdots, x_k)$ while the separation quadrance of any two points in $\mathcal{E}_k$ can be expressed in the form $q = \sum_{i=1}^{k} (x_i - y_i)^2$. We note the sum-of-squares representation of the number q.

We can view $\mathcal{E}_k$ as an infinite k-ball centred on an origin $(0, \cdots, 0)$ decomposable into sets of concentric k-spheres (*spheres*) each consisting of sets of lattice points with integer-valued co-ordinates and of constant quadrance. That is, each sphere of quadrance n consists of that set of lattice points that sit precisely on the surface described by the Diophantine equation $n = x_1^2 + \cdots + x_k^2$ and each (closed) *ball* of quadrance n consists of the set of all lattice points located on all spheres in $\mathcal{E}_k$ of quadrance $q \leq n$.

2. $\mathcal{E}_k$ AS TOPOLOGICAL SPACES

$\mathcal{E}_k$ spaces are discrete topological spaces [2] that we can regard as infinite sets of lattice points with integer co-ordinates. Every subset of $\mathcal{E}_k$ is also a topological space so we regard spheres and sets of spheres concentric with the origin, of indices $i \in 0, 1, 2, \cdots$ say, as sub-spaces of $\mathcal{E}_k$.

Suppose that the automorphisms of a particular representative lattice point (that is, all the signed permutations of the co-ordinates) of the i^{th} sphere at radius r_i in $\mathcal{E}_k$ define a set of points, this is a subset of the sphere that we will call a *shell*. The nature of this definition via automorphism assures the uniform distribution of lattice points of the shell over a sphere and thus the essential spherical symmetry of the shell. Note that while a shell does not necessarily contain all the lattice points of a sphere, any point on the sphere is necessarily an element of a shell. The consequential decomposition of a sphere into shells constitutes a partition of the set of lattice points that makes up the sphere [3]. Since the quadrance $q_i = r_i^2$ of a shell in $\mathcal{E}_k$ corresponds to some radius $r_i \in \mathbb{R}^+$, each multiplication of r_i by $m \in \mathbb{N}$ corresponds to a notional transformation of the shell as $q_i \mapsto m^2 q_j$ for some $m \in \mathbb{N}^2$. Such radial translation leaves both the order of the shell (the number of lattice points in the shell) and the pattern of dispersion of lattice points for any component shell of the sphere invariant, we will refer to the particular radially invariant pattern of this dispersion as the *point topology* of the shell and extend the terminology to spheres.

In $\mathcal{E}_k$, balls spheres and shells, well-defined sets of lattice points exhibiting spherical symmetry, will be treated as entities. The $m^2q_i,\ m \in \mathbb{N}$ thus specify the quadrances of the component shell entities of a *shell space*, another topological subspace of $\mathcal{E}_k$.

3. Sub-spaces of $\mathcal{E}_k$ as domains of group actions

It is clear that a shell space in $\mathcal{E}_k$ forms a group under radial translation of any distinguished shell, in that for any $m \in \mathbb{N}$ and shell quadrance q_i, under integer radial multiplication $q_i \mapsto m^2q_i$.

We note that if the quadrance q of a shell in a shell space is not square-free then we can find a square-free version of q that specifies a *primitive* shell in the shell space, the lowest shell of that topology in $\mathcal{E}_k$. Whenever the co-ordinates of a representative lattice point of a shell are not relatively prime then there must exist a shell lower (that is, of lesser quadrance) in the shell space; where the co-ordinates are relatively prime then the shell is primitive. Every primitive shell of quadrance q then generates the entire shell space of of spherically symmetrical lattice points radiating out from the origin. We note that as a consequence of the discreteness of $\mathcal{E}_k$, shell spaces are inherently disconnected from each other.

4. $\mathcal{E}_k$ as solution spaces for Diophantine equations

An isomorphism between solution spaces of Diophantine equations $n = x_1^2 + \cdots + x_k^2$ in squared integer variables and Euclidean geometric spaces will allow of a general expression for the number of ways in which numbers $n \in \mathbb{N}$ can be represented by sums of k-squares, which enumeration corresponds directly to the numbers of lattice points on spheres in $\mathcal{E}_k$.

4.1. **Partitions of numbers into squares of integers.** The classical definition of a partition of a number by Andrews [4] is of a non-increasing sequence of positive integers, the *parts* (of the number), that sum to the number. We make a new definition 4.1 that is both restricted and extended:

Definition 4.1. *A k-partition of a number $n \in \mathbb{N}$ is a non-ascending sequence of k integer square parts which sum to n.*

For example, $(16, 1, 1)$ and $(9, 9, 0)$ are both 3-partitions of 18. For brevity in this and derivative papers any reference to the term *partition* will always imply a k-partition.

We suppose that a number n has precisely $p_k(n) \geq 0$ k-partitions. For $p_k(n) > 0$ we sum the parts x_i^2 of a particular partition as:

$$n = x_1^2 + \cdots + x_k^2 \qquad (x_1^2 \geq x_2^2 \cdots \geq x_k^2 \geq 0) \tag{1}$$

and enumerate partitions arbitrarily as p^j where $1 \leq j \leq p_k(n)$.

For a given value of n and a k-tuple $X_k = (x_1, \cdots, x_k)$ of integers that satisfies (1), the automorphisms of X_k are also solutions to (1); we will refer to the solution space for all $n \in \mathbb{N}$ and all k-partitions of n as $\mathcal{N}_k$.

We note here that the validity of Theorem 4.2 which will establish an isomorphism that obtains between the solution set space and geometrical space is dependent on allowing the inclusion of parts of zero magnitude in the k-partitions of a number as in Definition 4.1.

Theorem 4.2. *The solution set space $\mathcal{N}_k$ is isomorphic to the Euclidean space $\mathcal{E}_k$.*

Proof. Every set of integers $(x_1, \cdots, x_k)$ that corresponds to a solution to (1) for a particular value of n can define the k co-ordinates of a unique lattice point in $\mathcal{E}_k$, that is, $\mathcal{N}_k \subset \mathbb{Z}^k$. Also, for every integer $y_i \in \mathbb{Z}$ the k-tuple $(y_1, \cdots, y_k)$ represents a unique solution to (1) for a value of $n \in \mathbb{N}$, so $\mathbb{Z}^k \subseteq \mathcal{N}_k$. The equality function $x_i = y_i$ that relates individual values in $\mathcal{N}_k$ to individual co-ordinates in $\mathcal{E}_k$ is then bijective. $\square$

The space $\mathcal{N}_k$ is dense in $\mathcal{E}_k$, every lattice point in $\mathcal{E}_k$ represents a unique solution in integers to a representation of $n \in \mathbb{N}$ as a sum of k squares of integers. If the variables in (1) had been real, the equation would have represented a sphere (the term taken to mean a k-sphere, including all dimensions $k \geq 1$) in real k-dimensional Euclidean space. As a Diophantine equation in integer variables the totality of solutions to (1) define a sphere of lattice points in the topological space $\mathcal{E}_k$. A *sphere* is taken to be *complete* if and only if its component lattice points represent all solutions to (1) for a given value of n.

We can regard any k-tuple as having a quadrance via (1); the signed permutations of a specific k-tuple representing sets of quadrance invariant points on a sphere are the lattice points of shells as previously identified. Such shells consist of mutually disjoint sets of lattice points on the sphere and their union, taking into account all the k-partitions of n, completes the sphere at quadrance n. Thus the set of all such shells constitutes a decomposition of the sphere with all component sets of lattice points interspersed on the sphere. For the j^{th} partition p^j of n we enumerate the number of corresponding solutions to (1) as $s_k^j(n)$, this is the number of lattice points in the j^{th} shell of the sphere. We immediately see the correspondence between a partition [3] of the set of lattice points on a sphere of quadrance n and a k-partition of the number n.

For example, while the automorphisms of the tuple $(3,3)$ define 4 lattice points in $\mathcal{E}_2$, namely $(3,3),(3,-3),(-3,3),(-3,-3)$, those of the 3-tuple $(3,3,0)$ define 12 lattice points in $\mathcal{E}_3$. In both spaces the sets of lattice points correspond to solutions to (1) which lie on spheres (planar in the first case, three-dimensional in the second) whose quadrance is $3^2+3^2=18$. There are two partitions of 18 in $\mathcal{E}_3$, namely $(9,9,0)$ and $(16,1,1)$.

Theorem 4.3. *Every solution in $\mathcal{N}_k$ to an instance of (1) is also a solution in $\mathcal{N}_{k+1}$.*

Proof. If a k-tuple in $\mathcal{N}_k$ be augmented by a co-ordinate of zero magnitude, a $(k+1)$-tuple is created. This represents a solution in $\mathcal{N}_{k+1}$ space of the same quadrance. □

For a given k-partition of a number n, since the main sequence $(x_1^2,\cdots,x_k^2)$ is by definition ordered, any elements of identical magnitude will be adjacent. We identify sub-sequences, each such containing only and all parts of the same magnitude. Following Andrews' terminology [4] we decompose a main sequence into t sub-sequences each of one or more square parts as $(\lambda_1^{n_1},\cdots,\lambda_t^{n_t})$, where each exponent n_i represents rather than a power, an enumeration of n_i identical parts of magnitude λ_i. We also enumerate the numbers m_j of non-zero parts in the main sequence and note that $k=\sum_{i=1}^{t} n_i$ while $n=\sum_{i=1}^{t} n_i\lambda_i$.

4.2. **Sums of squares of integers.** Problems relating to sums of squares of integers can clearly be related to partitions of a number $n\in\mathbb{N}$ in squares of integers. The number of ways of representing $n\in\mathbb{N}$ as a sum of k squares of integers is traditionally designated $r_k(n)$[5], we will refer to the number of distinct solutions to (1) corresponding to the j^{th} partition p^j of n in k squares as $s_k^j(n)$.

Theorem 4.4. *The number of distinct solutions to* (1) *corresponding to the* j^{th} *partition* p^j *of* n *in* k *squares is given by* (2)*:*

$$s_k^j(n) = 2^{m_j} \frac{k!}{\prod_{i=1}^{t} n_i!}. \tag{2}$$

Proof. The number of distinct automorphisms of each k-tuple in a solution set depends both on the number of ways in which sign can be assigned to the individual co-ordinates and on the numbers of distinct parts in the partition of the number. The first is clearly 2 for every non-zero part, resulting in a factor of 2^{m_j}; the second is the number of distinct ways in which the co-ordinates of the k-tuple can be permuted. If all the co-ordinates are distinct then this latter number is $k!$ but for each sub-sequence of n_i repeated values, the net number of ways is reduced by a factor of $n_i!$. Hence the result in (2). □

Corollary 4.5. *The order of a shell (that is, the number of component lattice points) is* $s_k^j(n)$ *as given by* (2)*.*

Proof. From the isomorphism of Theorem 4.2. □

Corollary 4.6. *The number of ways of representing* $n \in \mathbb{N}$ *as a sum of* k *squares of integers is:*

$$r_k(n) = k! \sum_{j=1}^{p_k(n)} 2^{m_j} \prod_{i=1}^{t} \frac{1}{n_i!} \tag{3}$$

Proof. The total number of lattice points that lie on the sphere of quadrance n is equal to the total number of distinct solutions to the representation of n as a sum of squares for all $p_k(n)$ partitions of n. Since the number of lattice points in the sphere is equal to the totality of those in all the $p_k(n)$ shells we have:

$$r_k(n) = \sum_{j=1}^{p_k(n)} s_k^j(n), \tag{4}$$

and the result follows from (2). □

5. Decompositions of shells in $\mathcal{E}_k$

The j disjoint shells of a sphere of quadrance n in $\mathcal{E}_k$ each incorporates $s_k^j(n)$ lattice points and together the set of shells provides a meaningful decomposition of the sphere via well defined subsets of the $r_k(n)$ lattice points of the sphere. This decomposition provides significantly more useful information about the point topology of a sphere than more traditional analyses of $r_k(n)$ in terms of divisor sums. A simple and fast algorithm has been developed to provide the sphere and shell content information for any dimension k.

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